

# Andrei Tregubov

## Data Scientist · R&D Engineer

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[linkedin.com/in/andrei-tregubov-rnd](https://linkedin.com/in/andrei-tregubov-rnd) | [github.com/oster-field](https://github.com/oster-field) | [andrei-tregubov.works](https://andrei-tregubov.works)  
Born on 31 March 2001 in Nizhny Novgorod | Authorised to work full-time in Germany



## PROFILE

Data Scientist and R&D Engineer with **3+ years** of experience building end-to-end data pipelines and **production-ready machine learning models** on large-scale time series data. Expertise with proven impact in **signal processing** and probabilistic **risk modelling** applied to real-world data. MSc in Computational Mathematics, with a **strong mathematical foundation** and a proven ability to deliver robust, deployable software. **Author of peer-reviewed research** published in Springer journals.

## PROFESSIONAL EXPERIENCE

### Data Scientist & R&D Engineer

*Institute of Applied Physics* | Jul 2021 - Nov 2024 · Remote

*Worked in a small cross-functional research team, driving technical and analytical work.*

- **Independently designed and built** an end-to-end sensor [data processing application](#), reducing dataset processing time from 10 hours to 15 minutes (**40x speedup**), replacing fully manual, dataset-specific workflows with a fully automated solution covering **ingestion, quality control, feature extraction, and anomaly detection**
- **Engineered** a domain-driven **data modelling and processing methodology** to clean, segment, and extract representative statistical features from over **2.1 billion** time-series data points, enabling reliable **rare event analysis** from long-term field measurements
- Developed a **machine learning model for predictive risk assessment**, achieving a **52% detection rate** on rare hazardous events; subsequently **commercialised** to improve maritime operational safety
- **Formulated** a novel analytical [model](#) (discovered and proved a previously unknown mathematical theorem) for **probability estimation of rare events** in complex stochastic time-series data
- Secured a **personal research grant** (worth €130,000); presented findings at **19 international conferences**; results published in **2 Springer journals** (Scopus-indexed)

## EDUCATION

**MSc in Computational Mathematics** · University of Passau · Passau, Germany · Nov 2024 - Mar 2026 · **GPA 1.8**

**BSc in Fundamental Mathematics** · Higher School of Economics · Moscow, Russia · Sep 2019 - Jun 2023 · **GPA 1.6**

## SKILLS

**Programming languages:** Python, MATLAB, R, SQL

**Python libraries:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, XGBoost, Statsmodels, SQLAlchemy, Transformers

**Tools:** Git, Docker, AWS, Tableau, Linux, Jupyter, Flask, REST APIs, Claude Code

**Core expertise:** Machine Learning (ML Pipelines), Research, Mathematics, Data Analysis, Predictive Risk Modelling, Data Science, Automation, Statistics, Anomaly Detection, Time-Series Analysis, Signal Processing, Big Data, Hypothesis Testing, Correlation Analysis, Probability Theory, Numerical Methods, Scientific Computing, Data Visualisation, Simulation & Modelling

## RESEARCH & PUBLICATIONS

### Frequency Spectra and Surface Wave Height Distributions Based on Measurements off Sakhalin Island

*Atmospheric and Oceanic Physics · Springer (Scopus-indexed)* · Oct 2025 · DOI: [10.1134/S0001433825700884](https://doi.org/10.1134/S0001433825700884)

Large-scale analysis of real-world sensor data; statistical modelling of extreme event probability

### Probability Distributions for Finite Ensembles of Irregular Waves

*Springer Proceedings in Earth and Environmental Sciences (Scopus-indexed)* · Nov 2023 · DOI: [10.1007/978-3-031-47851-2\\_19](https://doi.org/10.1007/978-3-031-47851-2_19)

Numerical simulation and probabilistic modelling of anomalous events in time-series data

## LANGUAGES

German B2 · English C2 · Russian C2 (Native)



Munich, July 2026